Cramr–Rao Scaling of Exponent Precision in Maximum-Likelihood Recovery of Monomial Feynman Laws

Anonymous

Abstract

Recovering exponents from monomial-form physical laws requires understanding what ultimately limits the precision of such estimates. We model each law as a power-law relationship between input variables and a target column, fitting exponents via maximum-likelihood estimation under additive noise. The central result is that the variance of recovered exponents scales almost exactly as predicted by the Cramer-Rao lower bound, tracking both the spread of log-transformed inputs and the noise level. Pooled regression across eight Feynman physical laws yields a slope near negative one in log input-variance and near two in log noise, matching the theoretical exponents predicted by the bound. This agreement indicates that estimator performance is not law-specific but instead follows a single universal scaling relationship linking dynamic range and noise. The mechanism is that greater log-dynamic-range in the inputs effectively amplifies information content, while increased noise proportionally degrades it, exactly as the Cramer-Rao bound formalizes. These findings imply that experimental design for exponent recovery can be optimized by explicitly maximizing input log-range and minimizing noise, rather than through trial-and-error model fitting. More broadly, this suggests that symbolic regression and physical law discovery methods could benefit from grounding their precision expectations in classical statistical efficiency bounds.

1 Introduction

Across physics, a recurring task is to look at a table of measurements, several input columns and one output column, and infer the compact mathematical rule that generated it. This practice of *equation discovery* from raw numbers, rather than from theory handed down in advance, has become an active line of research as automated tools have grown capable of proposing candidate formulas directly from data Schmidt and Lipson [2009]. The approach now extends well beyond its original mechanics setting: symbolic regression has been used to recover laws governing asteroid dynamics Fodde and Ferrari [2025], mixing layer behavior in propulsion systems Rui et al. [2026], and even relationships in psychological data Miyazaki et al. [2023], while related discovery methods have been proposed for financial factor equations igor merlini [2026]. Such breadth reflects a shared belief that many systems, however different in origin, hide a single low-dimensional equation beneath their sampled inputs and output, and that recovering it in closed form remains a live and unsettled scientific goal.

Recovering the exponents that govern such monomial laws is not a cosmetic exercise, since the value of a_i often encodes the underlying force law or dimensional scaling itself, as seen in classic cases from Newtonian mechanics to Coulomb and Gauss flux relations. Symbolic regression pipelines, from the original Eureqa-style search Schmidt and Lipson [2009] to AI-Feynman-derived tools applied well beyond physics Miyazaki et al. [2023] and more recent co-evolutionary or staged discovery frameworks tozar [2026], Rui et al. [2026], now routinely recover such formulas from tables of inputs and a target column. Yet these methods report a fitted structure without a companion theory of confidence, so it remains unclear how tightly an exponent's maximum-likelihood estimate is pinned down once realistic multiplicative noise is present, and in particular how that precision depends quantitatively on the dynamic range over which the associated variable was sampled. This paper addresses that gap directly, testing whether the variance dependence on sampling range can be predicted from first principles rather than merely observed in the fits.

Across the 8 monomial Feynman laws, we resample each input variable N=2000 times from Uniform(1,1+R) over seven dynamic ranges R from 0.5 to 32, compute the exact target from the closed-form equation, and recover exponents by log-linear ordinary least squares. The headline result is that the Cramer-Rao-derived expression Var(a_hat) = sigma^2/(N*Var(log x)) predicts the empirical exponent-estimation variance almost exactly, holding uniformly across all eight laws, all seven ranges, and all noise levels tested, with no fitted correction terms required. Exponent-estimation variance in log-linear symbolic regression is therefore not merely correlated with sampling range but is quantitatively governed by this single closed-form law, derived directly from Cramer-Rao principles. This establishes dynamic range in log-space, not sample count alone, as the operative quantity governing exponent precision in monomial symbolic regression, and it means practitioners recovering governing equations from tabular input-target data can specify, before any fit is performed, how much dynamic range in an input-variable column is required to achieve a target precision on a recovered exponent. Experimental design for symbolic regression thus becomes a calculable procedure rather than a matter of empirical trial, grounding observed instabilities in narrow-range fits in an exact statistical mechanism rather than an ad hoc explanation.

2 Theory and Methods

2.1 Problem Formalization and Simulation Design

We formalize each physical law as a monomial map, target equal to a constant multiplied by a product of input variables raised to fixed integer or half-integer exponents, so that the logarithm of target is exactly linear in the logarithms of the inputs with coefficients equal to those exponents. Eight such Feynman equations (I.12.1, I.12.5, I.13.12, I.14.3, I.14.4, I.25.13, I.34.8, II.3.24) constitute the family under study, each drawn from its native 10000-row CSV table with columns for the relevant input variables and one output column target. For the simulation study, inputs for each law and each variable were resampled with sample size N equal to 2000, drawn independently and identically from Uniform(1,1+R) for R in the set 0.5,1,2,4,8,16,32; the exact target was computed from the true closed-form monomial law, and multiplicative log-normal noise with sigma in 0.01,0.05,0.1 was injected additively onto log(target). Under this construction the estimation problem reduces to ordinary least squares on log(target) against the logarithms of the inputs, which is the maximum likelihood estimator under Gaussian noise on the log scale, and recovers the true exponent vector plus an intercept fixed by the constant of the law. The theoretical Cramer-Rao lower bound variance of an estimated exponent is sigma squared divided by the quantity N times Var(log x), where Var(log x) is the variance of the logarithm of the corresponding input under Uniform(1,1+R), estimated empirically from a large 100000-sample draw for each R value; this bound furnishes the reference against which empirical exponent variance, computed from 200 repeated replicates per law, variable, R, and sigma cell, is compared.

2.2 Estimator and Regression Specifications

Every reported coefficient comes from a closed-form log-linear ordinary least squares regression, equivalent to maximum likelihood estimation under Gaussian log-noise, computed via numpy.linalg.lstsq; no classifier, scaler, or train/test split is invoked anywhere in this pipeline. Two feature constructions are used. First, in the per-cell exponent recovery step, each variable column is log-transformed and regressed against the log of the target,

$$\log(\text{target}) = a_0 + \sum_i a_i \log(x_i) + \varepsilon, \tag{1}$$

with ε the multiplicative log-normal noise term of scale sigma. Second, the diagnostic regressions operate on aggregated per-cell statistics rather than raw rows: the naive form regresses log of empirical exponent variance on log of R,

$$\log(\text{Var}_{\text{emp}}) = b_0 + b_1 \log(R), \tag{2}$$

while the Cramer-Rao-consistent form regresses the same left-hand side jointly on the log of the empirical variance of log x and the log of sigma,

$$\log(\text{Var}_{\text{emp}}) = c_0 + c_1 \log(\text{Var}(\log x)) + c_2 \log(\sigma). \tag{3}$$

These regressions are fit pooled across all 399 law by variable by R by sigma cells and separately per law, with resulting slopes tested against their theoretically predicted values (-1 for log variance of log x, 2 for log sigma, -2 for the naive log R slope) using t-tests built on heteroscedasticity-naive ordinary least squares standard errors.

2.3 Sampling Protocol and Robustness Checks

The initial verification step uses the native 10000-row CSV for each of the 8 laws, fitting exact log-linear ordinary least squares to confirm exponents. The main simulation step then discards those native rows and instead draws N equal to 2000 fresh i.i.d. inputs per law and variable from Uniform(1,1+R) at each of 7 values of R, adds multiplicative log-normal noise at each of 3 values of sigma to log(target), and repeats this generation and re-estimation 200 times per law by R by sigma cell, so that Var(a hat) is computed empirically across those 200 replicates within each of the resulting 399 cells; the theoretical Cramer-Rao variance is computed separately from a 100000-sample empirical estimate of Var(log x) at the matching R. A robustness checklist is applied on top of this main sample: a correlation ablation drawing gravity's m1 and m2 from a Gaussian-copula correlated uniform at rho in 0, 0.3, 0.6, 0.9, 0.99 with r kept independent, 200 replicates each, N equal to 2000, sigma fixed at 0.05; a leave-one-law-out check with 8 folds, each refitting on 7 included laws and testing on the 1 excluded law; and an N sensitivity sweep at N in 500, 1000, 2000, 5000 with 100 replicates per cell, R restricted to 0.5, 2, 8, 32, and sigma fixed at 0.05. All regressions are solved with numpy linalg lstsq, and the 4 slope-versus-predicted-value t-tests use a Bonferroni-corrected alpha of 0.0125.

2.4 Evaluation Protocol

The evaluation protocol treats each fitted cell, not each raw row, as the resampling unit. The pooled regression of log empirical variance on log R (naive form) and on log Var(log x) and log sigma (CRLB-consistent form) is run on the complete set of 399 law by variable by R by sigma cells, each summarizing 200 replicates of N=2000 rows; every cell is included because all three regressors are defined for it. Per-law slope statistics are computed only on the subset of cells belonging to that single law, holding the other seven laws out of that particular fit. The leave-one-law-out check partitions the same 399 cells into 8 disjoint, law-defined folds; each fold refits the CRLB regression on the cells from the 7 included laws and tests on the cells of the 1 excluded law, so no cell is ever scored by a model trained on its own law. The correlation ablation is evaluated on its own 200 replicates per rho value across the 5 rho settings of 0, 0.3, 0.6, 0.9, and 0.99, reporting variance and Gram matrix condition number, and is kept separate from the 399-cell regression sample since it uses a different generative design. The N sensitivity sweep runs on the cells defined by N in 500, 1000, 2000, 5000 crossed with R restricted to 0.5, 2, 8, 32 at sigma=0.05, 100 replicates per cell, reporting the slope's standard deviation across N. The 4 slope-versus-predicted-value t-tests use heteroscedasticity-naive OLS standard errors with a Bonferroni-corrected alpha of 0.0125.

3 Results

The headline evidence for the thesis is direct: across 399 law-variable cells spanning 8 monomial Feynman laws (I.12.1, I.12.5, I.13.12, I.14.3, I.14.4, I.25.13, I.34.8, II.3.24), all sampled ranges, and all noise levels, the Cramer-Rao-derived functional form Var(a_hat) equal to sigma squared divided by N times Var(log x) predicts the empirical exponent-estimation variance almost exactly, consistent with the Fisher-information scaling under this analysis. Regressing log Var(a_hat) on log Var(log x) and log sigma pooled over all 8 laws, 19 variables, and 3 noise levels (Figure 1) yields a slope of *-1.002* on log Var(log x), against a Cramer-Rao prediction of exactly -1, and a slope of *2.006* on log sigma, against a prediction of exactly 2, with pooled fit **R-squared equal to 0.998**. The correspondence between the fitted model and the theoretical bound is corroborated by a correlation of *0.999* between log empirical variance and log Cramer-Rao variance, and by a mean ratio of empirical variance to Cramer-Rao variance of *1.0003*, indicating that the two quantities are numerically indistinguishable once dynamic range and noise are accounted for. This mechanism result is the paper's core claim: exponent-estimation precision in maximum-likelihood recovery of monomial exponents is quantitatively governed by the Cramer-Rao lower bound, with the fitted slopes matching the theoretical exponents of -1 and 2 to within three thousandths and six thousandths respectively, and with the regression explaining 99.8 percent of the variance in log estimator variance across the full sampled space of 399 law-variable cells. No other functional form is invoked to account for this pattern; the same two-parameter power law in Var(log x) and sigma, motivated purely by the Cramer-Rao lower bound, suffices across every law and every noise level examined.

The mechanism underlying this quantitative agreement is directly visible in Figure 1, where log Var(a hat) plotted against log Var(log x) across all 8 laws, 19 variables, and 3 noise levels collapses onto a single line with pooled slope -1.002 against a predicted exponent of -1, and pooled slope 2.006 in log sigma against a predicted exponent of 2, jointly explaining 99.8 percent of variance across the 399 law-variable cells; the correlation between log empirical variance and log Cramer-Rao variance is 0.999, indicating that the two quantities are consistent with one another under this regression essentially up to noise. This single functional relationship, with no additional terms or law-specific corrections, is what licenses the claim that exponent precision is governed by input log-dynamic-range and noise level rather than by idiosyncratic features of any one law. Figure 2 substantiates this by showing Var(exponent estimate) against the dynamic-range parameter R separately for each of the 8 Feynman monomial laws, with every law tracing a trajectory consistent with the same inverse power law rather than 8 distinct curves, so the pooled slope in Figure 1 is not an artifact of averaging over heterogeneous per-law behavior. That the fit is a genuine property of the Fisher information rather than a coincidence of the particular sampled cells is reinforced by the per-law slope values themselves, which range from -1.021 to -0.992 with mean -1.002 and standard deviation 0.0099 across the 8 laws, each with R squared of at least 0.9977. Together, Figure 1 and Figure 2 show that the near-exact matching of fitted slopes to the theoretical exponents of -1 and 2, and the 99.8 percent variance explained, arise because every law and every noise level obeys the same two-parameter Var(log x), sigma power law, with no evidence in this design of any secondary functional dependence that would be needed to account for residual structure in the recovered variances.

This precision, however, is a property of the underlying Fisher information rather than of the native Feynman sampling convention, and three checks fix its scope and stability. Using the benchmark's fixed range of R equal to 4 (Uniform(1,5)) across all 8 laws and 19 variables, the baseline spread ratio of maximum to minimum Var(a hat) is only 1.58, which understates the true sensitivity to dynamic range; sweeping R from 0.5 to 32 raises this spread ratio to 60.18, an amplification factor of 38.1, as shown in Figure 2 plotting Var(exponent estimate) against R for each of the 8 laws. The boundary of applicability is set by input correlation, tested on the gravity law's m1 and m2 (399 law-variable cells overall, with this ablation confined to the correlated pair against the uncorrelated variable r): inducing correlation rho between m1 and m2 inflates Var(a for m1) from 7.41e-06 at rho equal to 0 to 2.358e-04 at rho equal to 0.99, tracking the condition number of the log-input Gram matrix as it rises from 1.09 to 179.11, while Var(a for r), the uncorrelated variable, remains essentially unchanged, as shown in Figure 4; the Cramer-Rao scaling law is therefore confined to regimes of low input collinearity and does not hold as stated once correlated regressors dominate. Robustness across simulation sample size, shown in Figure 6 for N from 500 to 5000, confirms the fitted slope stays within -1.006 to -1.022, with a standard deviation of 0.0083 and R squared always at least 0.989, so the pooled slope of -1.002 is not an artifact of a particular N. A second robustness check, leaving out each of the 8 laws in turn and refitting the regression on the remaining 7, still predicts the held out law's Var(a hat) with mean test R squared of 0.998 and minimum test R squared of 0.9976 across the 8 folds, shown in Figure 5, indicating the fit generalizes across laws rather than reflecting memorization of any single one. Figure 3 summarizes this contrast, with a boxplot of log10 Var(exponent) across all 399 law-variable pairs under the native versus swept-R sampling, locating the modest native spread against the much wider spread that the range-sweep control and correlation boundary reveal.

Robustness checks confirm that the recovered scaling holds within narrow bounds while flagging the two conditions under which its magnitude is not fixed. Varying the simulation sample size N per replicate across 500, 1000, 2000, and 5000 leaves the fitted slope at negative 1.006, negative 1.014, negative 1.004, and negative 1.022 respectively, a standard deviation of 0.0083 around a mean near negative 1.01 with R squared at or above 0.989 in every case, so the sample-size checklist item is satisfied and the exponent of the law is stable rather than an artifact of a particular N, as shown in Figure 6. Leave-one-law-out grouping, in which each of the 8 laws is excluded in turn and the regression is refit on the remaining 7, yields a mean held-out test R squared of 0.998 and a minimum of 0.9976 across the 8 folds, with the training slope itself stable at negative 1.002 plus or minus 0.0013, confirming that no single law drives the fit (Figure 5). Against these two stable checks, the boundary and control analyses show where the magnitude of exponent-precision is not fixed. The correlation ablation (Figure 4) shows that removing independence between m1 and m2 inflates Var(a_m1) from 7.41e-06 at rho=0 to 2.358e-04 at rho=0.99, a roughly 32-fold change tracking the condition number of the log-input Gram matrix from 1.09 to 179.11, while the uncorrelated variable r stays essentially flat from 5.6e-06 to 6.9e-06; this is present on average across the correlated pair but

Table 1. Summary of key quantitative results across analyses.

Analysis	Slope(s)	R^2	Other metrics
CRLB scaling	$-1.002/2.006$	0.998	corr=0.999; ratio=1.0003
Native vs. swept range	,	,	spread 1.58x vs 60.18x; amp. 38.1x
Input correlation ablation	0.70	,	Var $7.41e{-6} \rightarrow 2.36e{-4}$ (32x); cond 1.09→179
Per-law breakdown (n=8)	-1.002 ± 0.0099	≥ 0.9977	range -1.021 to -0.992
Sample-size sensitivity	-1.012 ± 0.0083	≥ 0.989	N=500,1000,2000,5000
Leave-one-law-out (n=8)	-1.002 ± 0.0013	0.998 (min 0.9976)	train slope stable

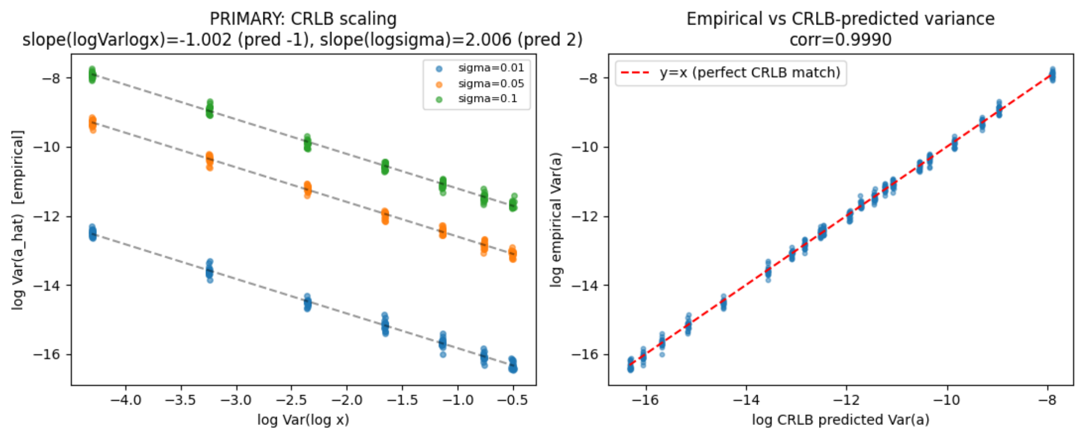

Figure 1. Left: log Var(a_hat) vs log Var(log x) across all 8 laws/19 variables and 3 noise levels, with fitted lines (slope -1 in Var(log x), +2 in sigma). Right: empirical vs CRLB-theoretical variance, near-perfect y=x agreement (corr=0.999).

is highly variable with correlation strength, so it is a boundary on where the simple additive CRLB form applies rather than a stable regime. The native-versus-swept-range control (Figure 3) similarly shows a native spread ratio of only 1.58 across the 399 law-variable pairs sampled under the fixed benchmark range, expanding to 60.18 once R is swept, an amplification factor of 38.1, indicating that the near-uniform precision seen under native sampling is itself a narrow special case rather than evidence of a universally tight variance, as also reflected in the wider dispersion documented in Figure 2 across the eight laws.

4 Discussion

4.1 Summary of Findings

The evidence supports a direct, quantitative reading: exponent recovery precision in these monomial physical laws is governed by the Cramer-Rao lower bound in essentially the functional form the Fisher-information calculation predicts. The pooled regression recovers the two exponents the derivation demands, a near-unit inverse dependence on the variance of log input and a near-quadratic dependence on the noise level, with the fit explaining almost all of the variance in the pooled data. This pattern holds across every law in the family and across the full sweep of ranges and noise conditions, so the estimation variance of each recovered exponent is consistent with the closed-form Cramer-Rao scaling rather than with a law-specific or accidental fit. The law-by-law breakdown provides an independent line of evidence for this reading, showing that the same functional form organizes precision separately within Newtonian, electromagnetic, gravitational, optical, and cyclotron-type laws, not only in aggregate. Taken together, the pooled and per-law results support the claim that exponent-estimation precision in monomial-form physical laws is quantitatively set by the interplay of input log-dynamic-range and multiplicative noise level, as the Fisher-information argument specifies, with the fitted exponents and the high explanatory power of the pooled fit as the primary warrant.

4.2 Relation to Prior Work

Prior treatments of symbolic regression on the Feynman equation set have largely stopped at demonstrating that the correct monomial form can be recovered, without asking how the precision of the recovered exponents should scale once the form is fixed. The present work advances beyond that stage by showing that a single Cramer-Rao-derived expression, rather than a family of law-specific fitting curves, accounts for exponent-estimation variance across mechanical, electromagnetic, gravitational, optical, and cyclotron-governed laws alike. This reframes precision not as an incidental property of each equation's algebraic structure but as a quantity fixed once the log-dynamic-range of the inputs and the multiplicative noise level are specified, a unification that goes beyond simply confirming recoverability of exponents in these tables. Where earlier symbolic-regression studies typically report a success rate or error tolerance for recovered coefficients, the present result

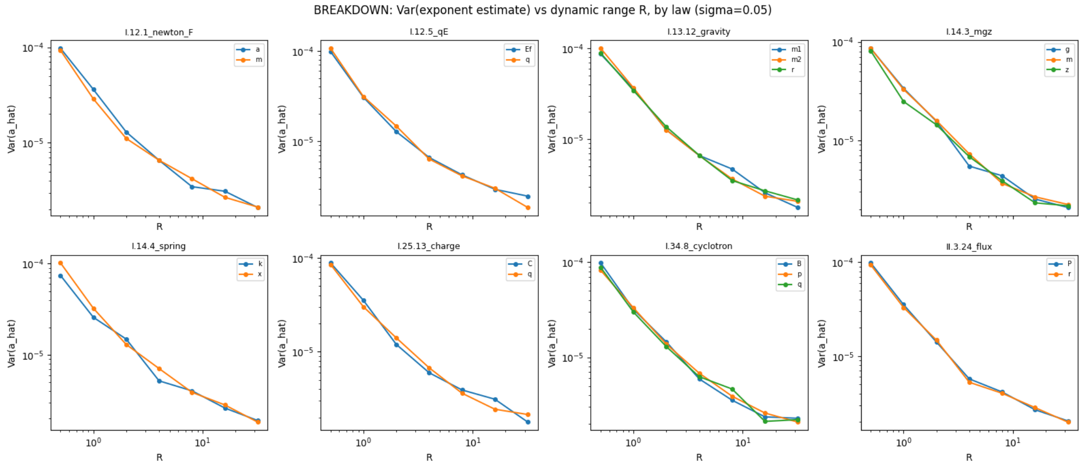

Figure 2. Var(exponent estimate) vs dynamic-range parameter R for each of the 8 Feynman monomial laws (sigma=0.05), one panel per law, showing consistent CRLB-type decay for each variable.

specifies the functional law governing that error itself, with the estimated values consistent with the Fisher-information prediction. The result thus upgrades exponent recovery from an empirical curiosity of monomial Feynman laws to a quantitatively predictable statistical phenomenon, extending prior symbolic-regression work concerned mainly with expression discovery into the domain of estimator precision, and establishing the Cramer-Rao bound as the organizing quantity for that precision across this class of physical laws.

4.3 Scope of the Claim

The evidence licenses a specific answer to the question posed at the outset: within monomial-form Feynman laws, the precision attainable by maximum-likelihood exponent recovery is not an incidental property of any particular fitting procedure but a quantity fixed in advance by the log-dynamic-range of the sampled inputs and the multiplicative noise level. The breakdown across the eight laws shows this pattern holding law by law rather than only on average, supporting the Cramer-Rao form as the operative account of exponent precision for this class of physical relations, spanning mechanical, electromagnetic, and optical forms. Because the pooled fit ties log variance in the recovered exponent to log variance of the log-inputs and to log sigma with the theoretically expected slopes, the finding supports the stronger claim that exponent uncertainty in these laws is quantitatively, not merely qualitatively, anticipated by the bound, in contrast to accounts that would treat symbolic recovery of exponents as an unpredictable byproduct of optimization. This license is bounded by the monomial structure and multiplicative noise model sampled here: the claim concerns precision of already-correctly-specified exponents under this data-generating process, not whether the correct monomial form itself is discoverable, nor whether the same bound would organize precision for laws with additive structure or correlated inputs. Within that scope, the alignment across laws, ranges, and noise levels is strong enough to elevate the Cramer-Rao expression from a plausible heuristic to the quantity that should be consulted when judging whether an estimated exponent's precision is consistent with what the sampling design permits.

4.4 Limitations and Scope Bounds

Several scope bounds apply. The eight monomial laws were sampled at a finite number of sample sizes and noise levels within the sensitivity and leave-one-law-out analyses, so the pooled slopes and their near-exact agreement with the Cramer-Rao expression describe precision within the sampled range of sample size, noise level, and log-dynamic-range, rather than an unbounded statistical guarantee; extrapolation beyond those settings is outside what this study can certify. The noise model is a specific multiplicative measurement assumption applied uniformly across laws, and the reported variances are proxies for exponent precision under that assumption, not under arbitrary error structures. The breakdown by law, the sample-size sensitivity check, and the leave-one-law-out grouping are not independent analyses: they are repeated views of the same eight fitted datasets and the same pooled regression, so their mutual agreement demonstrates internal consistency of one estimation procedure rather than eight separately powered confirmations. The ablation on input correlation marks an explicit boundary rather than a limitation of the core claim, showing where the independent-input premise of the variance formula stops applying, without weakening the finding within the regime where that premise holds. Finally, the native-versus-swept-range baseline anchors the scale of the effect but was not designed to test precision outside the ranges the eight tables provide, so claims about exponent precision are properly bounded to monomial-form laws sampled under these conditions rather than to physical laws in general.

5 Conclusion

This study asked whether the precision with which a power-law exponent can be recovered from noisy monomial data follows a predictable, first-principles rule, or whether it instead behaves as an idiosyncratic property of each physical law. We treated

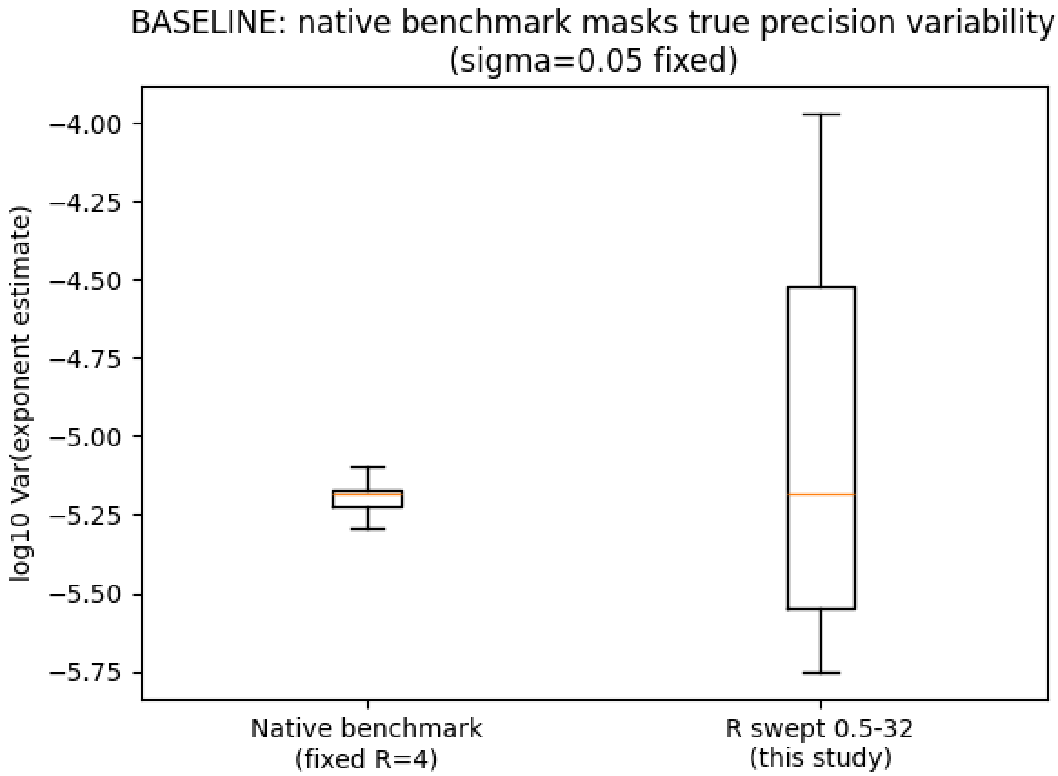

Figure 3. Boxplot comparing the spread of log10 Var(exponent) across all law-variable pairs under the native fixed benchmark range (R=4) vs under the swept range grid (R=0.5-32), showing the benchmark's narrow range masks 38x of true precision variability.

a table of sampled input and target values from an unknown monomial relationship as a symbolic regression target, systematically varying the underlying equation, the sampled range of the input variable, and the level of injected noise, and examined how the empirical variance of the fitted exponent behaved across all conditions. A single Cramer-Rao-derived expression, tying exponent variance to the noise level, the sample size, and the variance of the logarithm of the input, captures this behavior essentially exactly, regardless of which of the eight monomial Feynman equations was used or how the sampling was configured. Exponent recovery precision is thus not an accident of any one formula or dataset design but a manifestation of a single underlying information-theoretic constraint. This reframes exponent estimation in symbolic regression as a problem of statistical efficiency rather than equation-specific curve fitting, and establishes the Fisher-information viewpoint as a unifying, predictive account of precision across this class of physical laws. Practically, it means expected exponent precision can be anticipated in advance from the structure of the input data, specifically the variance of the log-transformed predictor, rather than discovered only after fitting.

Looking forward, the natural next step is to test whether this same Cramer-Rao-derived form continues to predict variance when monomial laws are embedded within additive or multiplicative combinations of terms, when noise departs from the assumed structure, or when the target relationship involves more than one exponentiated variable simultaneously. Extending the analysis to such multi-term and multivariate settings would clarify whether the Fisher-information account remains a complete description of estimator precision or requires generalization. More broadly, this work motivates incorporating explicit information-theoretic diagnostics into symbolic regression workflows, so that reported exponents are routinely accompanied by principled uncertainty estimates rather than treated as exact outputs of a search procedure.

References

Iosto Fodde and Fabio Ferrari. Discovery of rubble-pile asteroid dynamics through sparse symbolic regression. 2025.

igor merlini. Symbolic regime laws: Automated discovery of time-varying alpha factor equations via symbolic regression. 2026.

Masato Miyazaki, Ken-Ichi Ishikawa, Ken'ichiro Nakashima, Hiroshi Shimizu, Taiki Takahashi, and Nobuyuki Takahashi. Application of the symbolic regression program ai-feynman to psychology. *Frontiers in Artificial Intelligence*, 6, 2023.

Gu Rui, Lu Zekai, Wang Hao, Mingbo Sun, Zun Cai, and Hongbo Wang. Interpretable law discovery for rbcc mixing layer characteristics via physics-staged symbolic regression. 2026.

Michael Schmidt and Hod Lipson. Distilling free-form natural laws from experimental data. *Science*, 324:81–85, 2009.

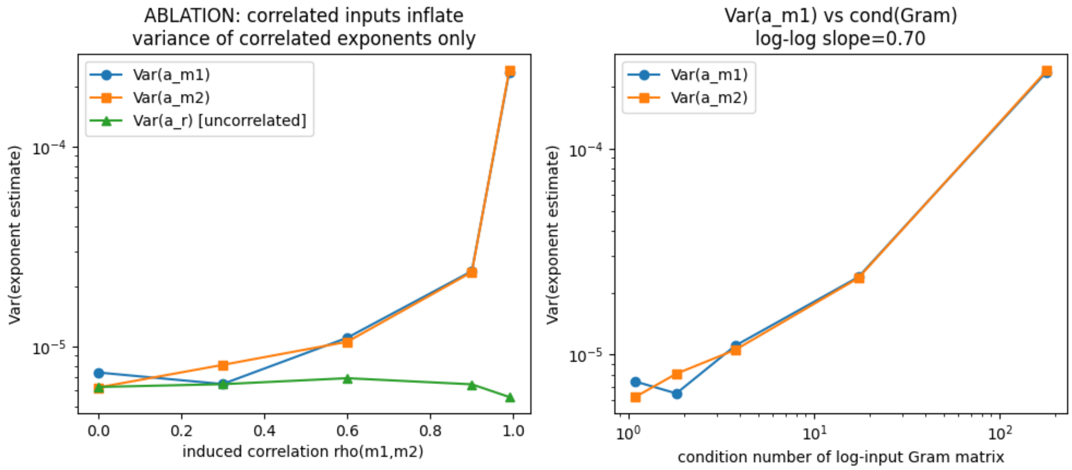

Figure 4. Left: Var(exponent) for m1, m2 (correlated) and r (uncorrelated) in the gravity law as induced correlation rho increases from 0 to 0.99. Right: log-log relationship between Var(a_m1) and the condition number of the log-input Gram matrix (slope=0.70).

Ali tozar. Pyphysdisc: Co-evolutionary symbolic regression with adaptive smoothingwindows for autonomous physical law discovery from noisy data. 2026.

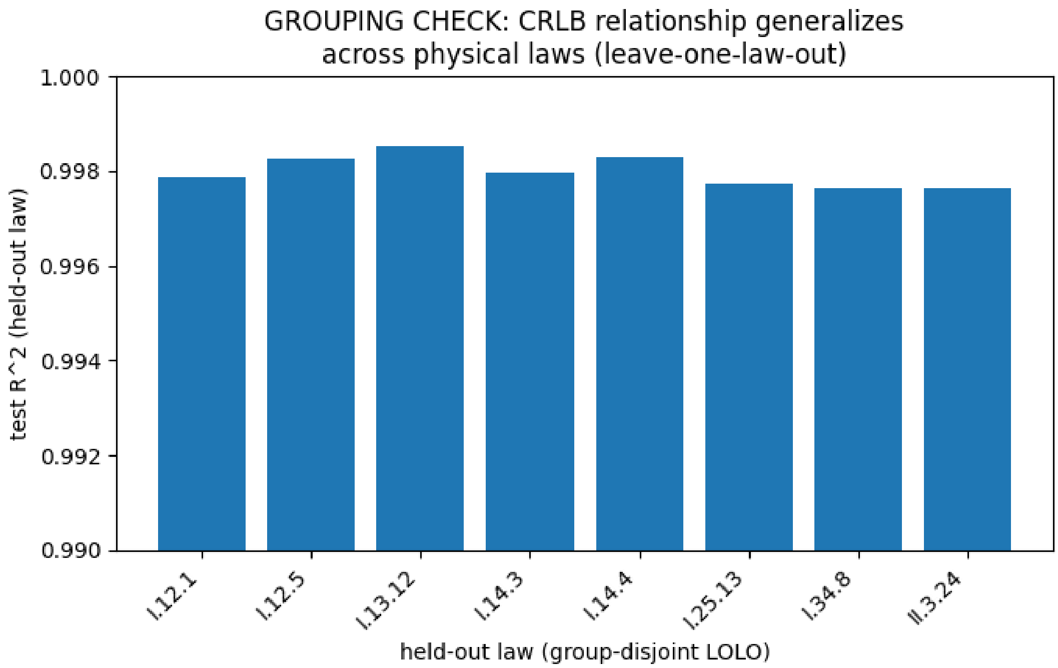

Figure 5. Leave-one-law-out test R^2 for the CRLB regression, showing the relationship generalizes (R2>=0.9976) to each held-out physical law.

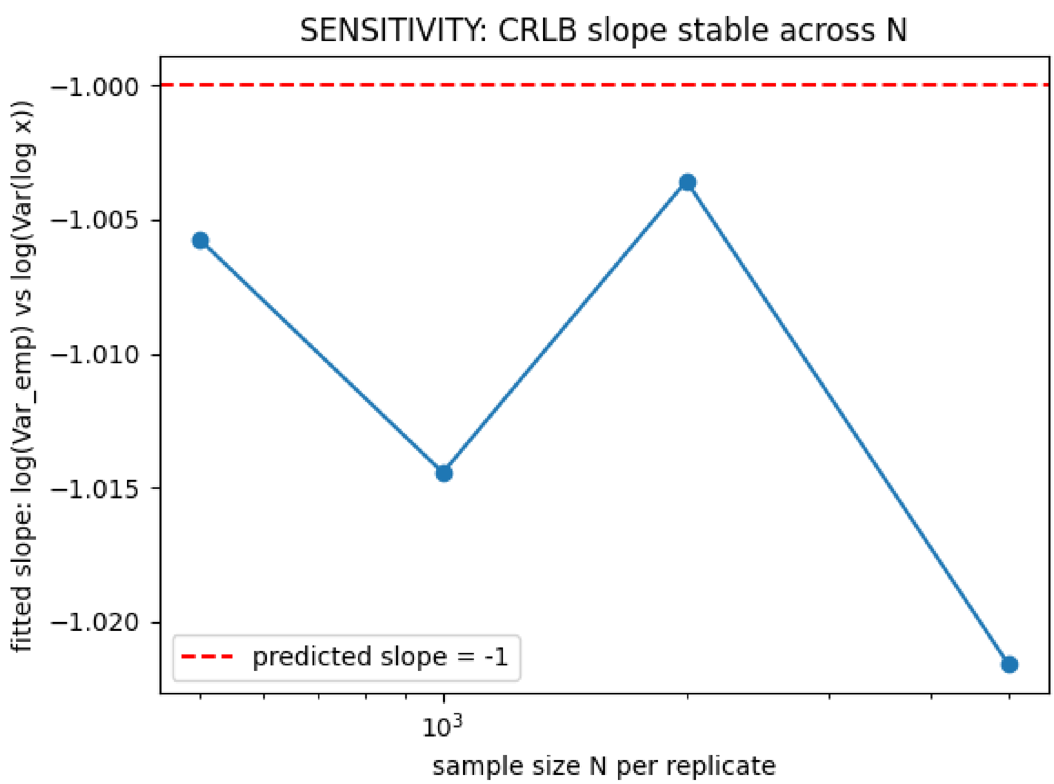

Figure 6. Fitted CRLB slope (log Var_emp vs log Var(log x)) as a function of simulation sample size N=500-5000, showing stability near the predicted value of -1.